Pattern Recognition

NYCU CS – Fall 2026

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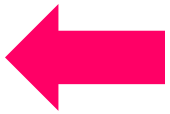
What is Pattern Recognition?

The goal, in the most general term, is to extract knowledge from data, and then to apply the knowledge to analyze (related) data.

The field of "pattern recognition" is only vaguely defined, and not everyone agrees on its exact scope. The following is given in one major textbook:

> *The field of pattern recognition is concerned with the **automatic** discovery of **regularities** in data through the use of computer algorithms and with the use of these regularities to take **actions** such as classifying the data into different categories.* (Bishop, 2006)

The term "pattern", according to different sources, can refer to

- The regularities in data
- Individual objects/items in data 

A Pattern Recognition Example

Q: What is this?



The next task: Identify the apple's variety.

(A) golden delicious **(B) red delicious**
(C) gala **(D) red rome**
(E) fuji **(F) McIntosh**
(G) granny smith

OK, you all know the answer (assumed)…

Now, what if you are given the following examples?


(A)


(B)


(C)


(D)


(E)


(F)

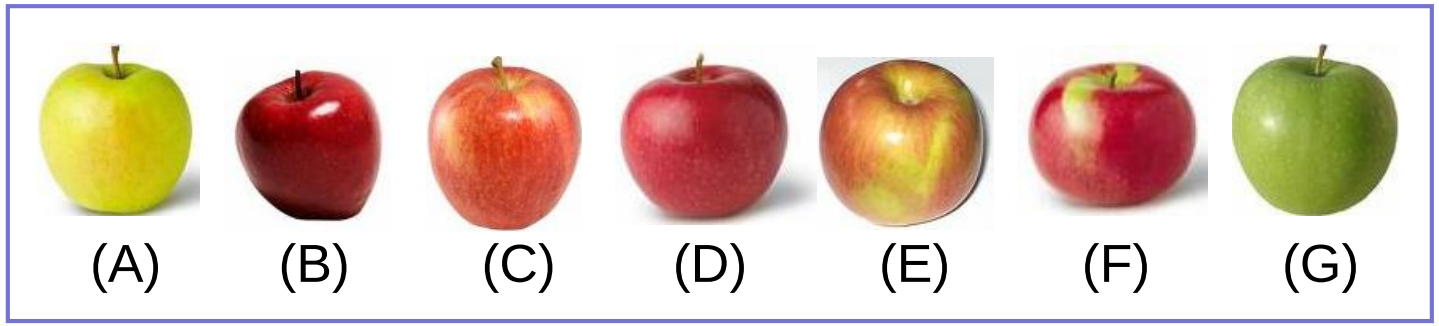

(G)

A Pattern Recognition Example

Now, how is this related to our course?

Actually, in your brain you have gone through the steps of a classification task in pattern recognition.

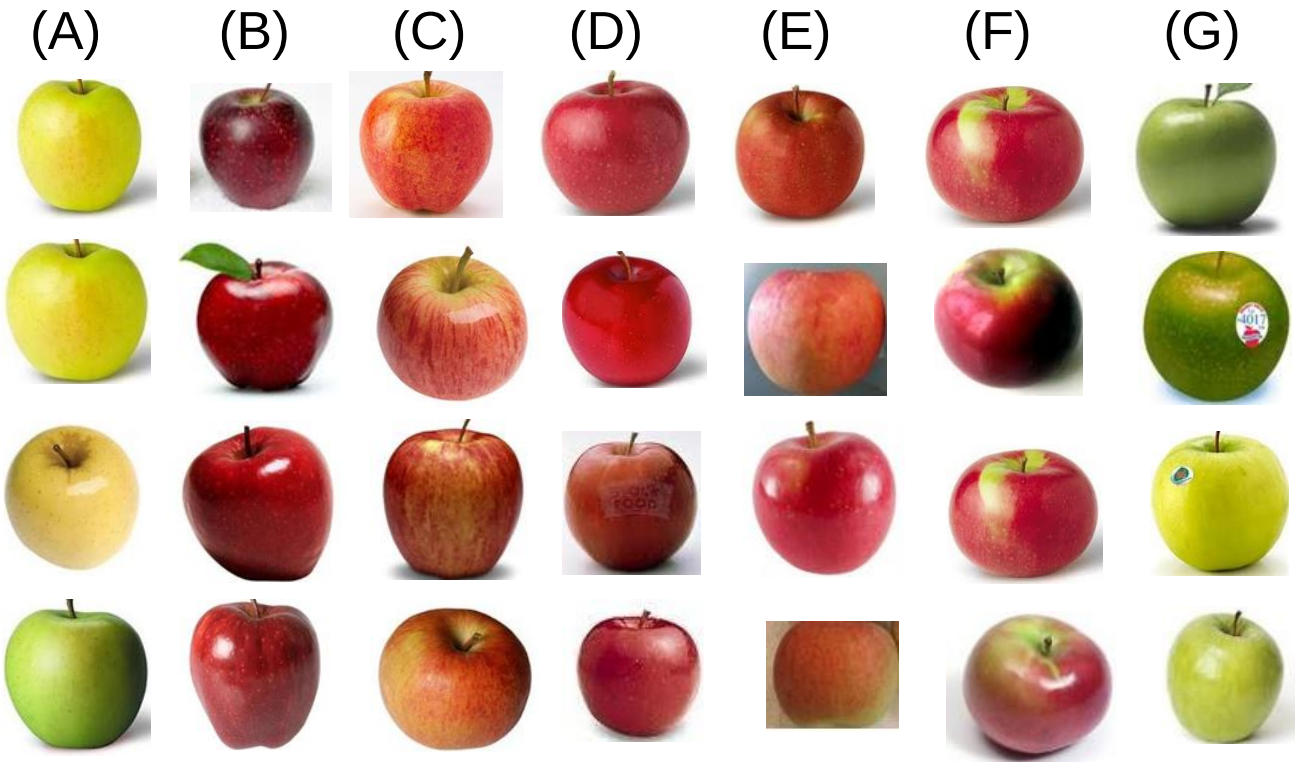
Our brains are doing tasks of pattern recognition all the time, and its part of our intelligent functions.



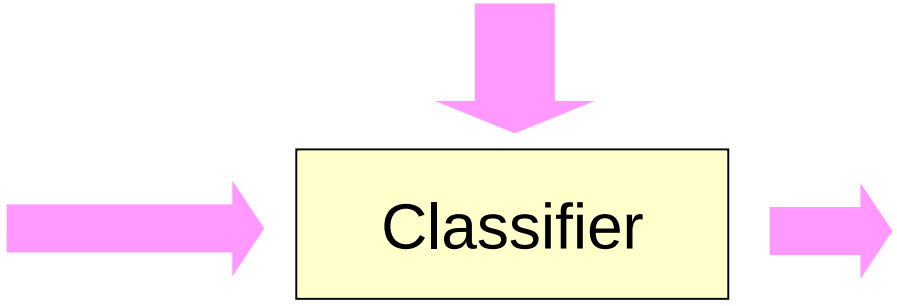
A Pattern Recognition Example

Consider:

- How you organize the information you learned from the training data, and
- How you use what you have learned to do the classification.

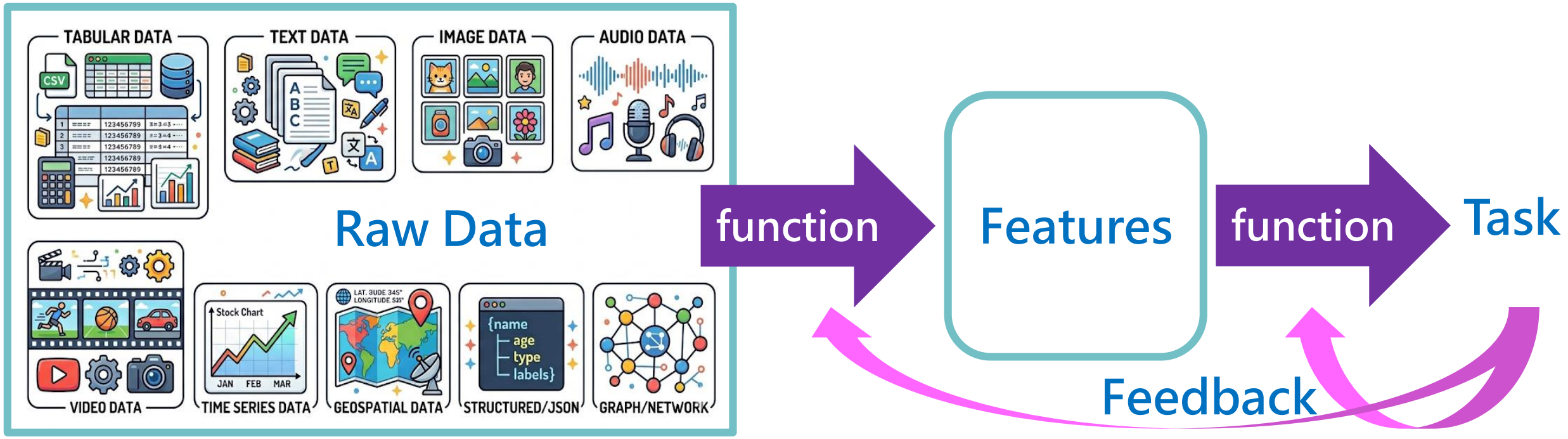






Pattern Recognition Overview

- What information do you use about the objects (apples) in order to make the decision? (In other words, what are the **features**?)
- How do you organize the information about the training data internally? (The design and training of a **classifier model**)



Pattern Recognition Overview



This course will put more focus on this part: How to represent data so the PR tasks (classification, etc.) become straightforward.

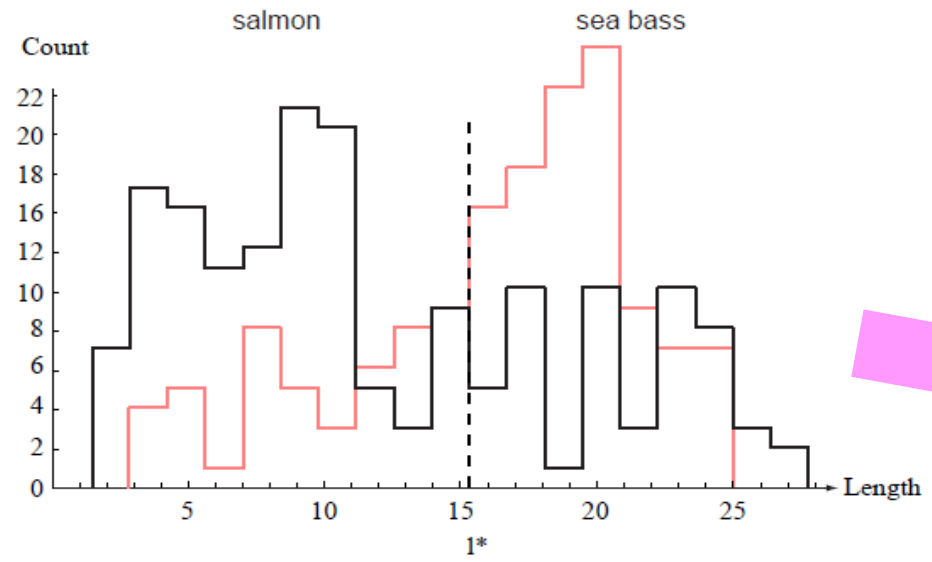
Traditional PR courses focus on this part by introducing various classification methods: Linear and logistic regression, MLP, decision tree and random forest, boosting, SVM, CNN, transformer, …

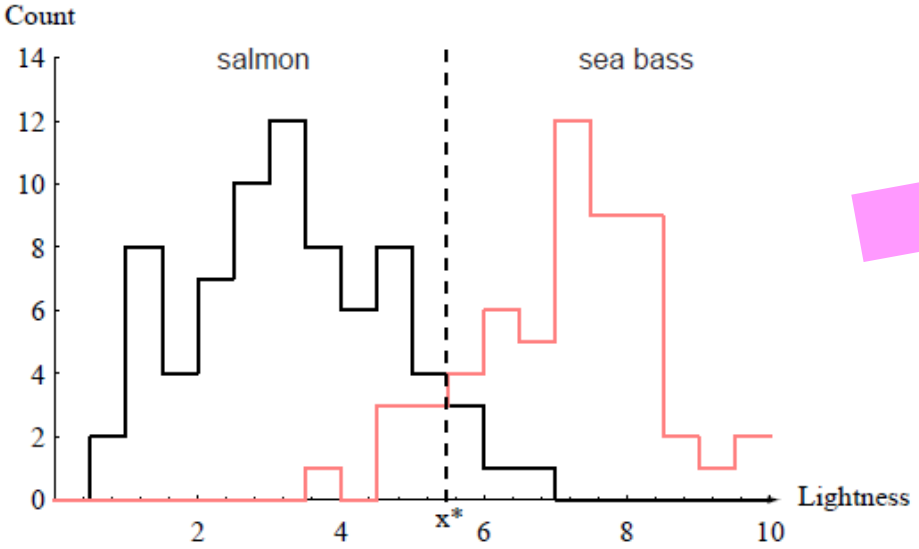
Example: Features and Feature Space

This is an example from the DHS book: To classify a fish as a sea bass or a salmon.

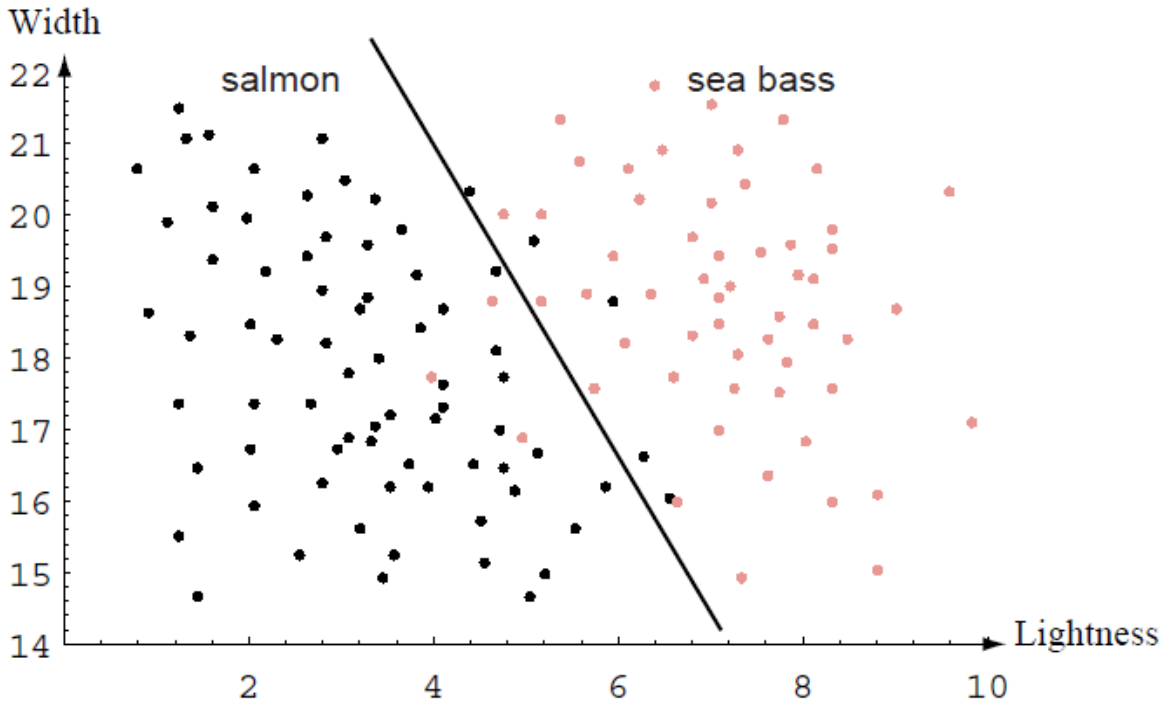






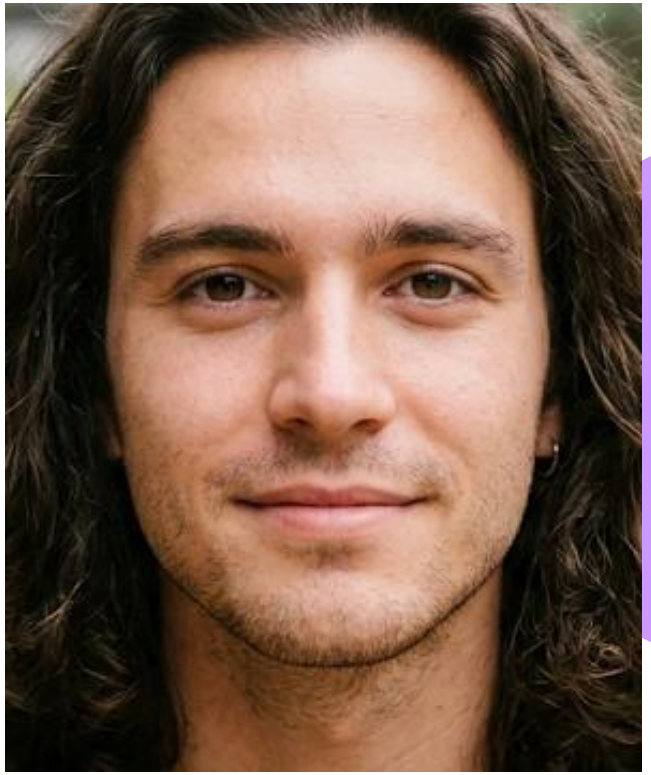


Feature Space



Example: Features and Feature Space

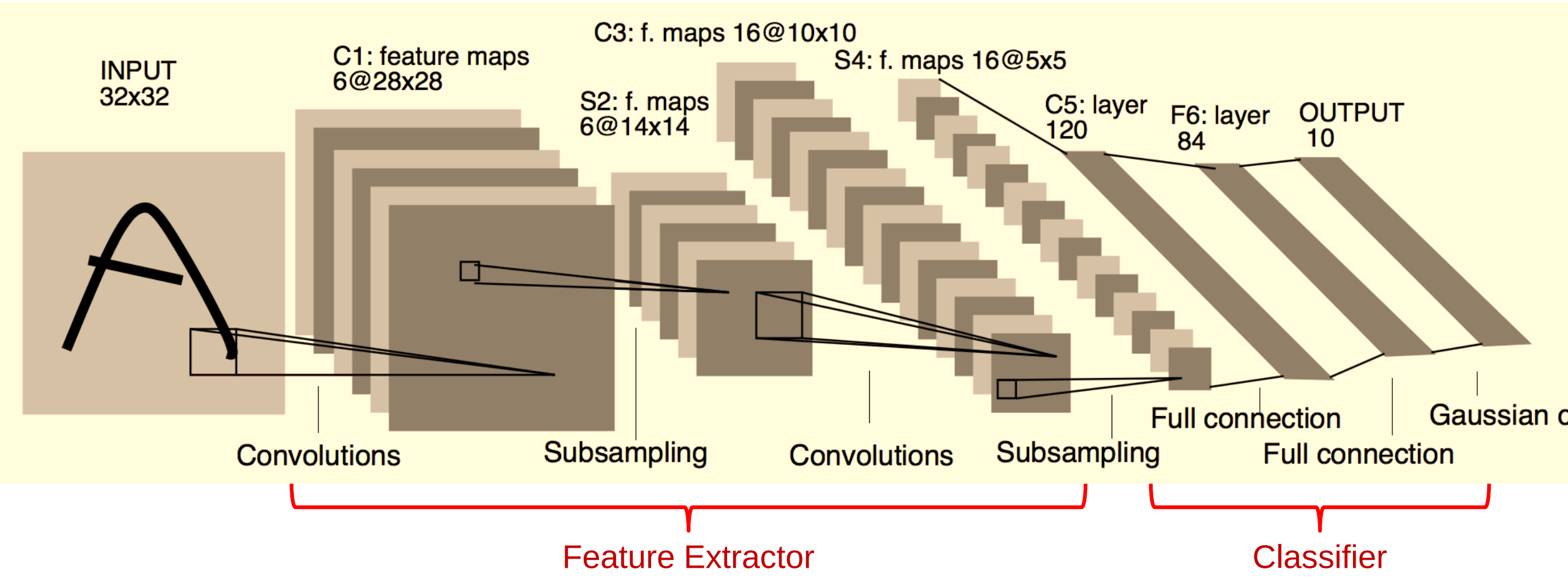
"Expert features" are not available for most problems. Example:



Male?
Female?
Why?



LeNet: A Basic Convolutional Neural Network



"You can think of a neural network as a complex geometric transformation in a high-dimensional space, folded and twisted to untangle the data... Essentially, a deep learning model is a complex geometric transformation that unravels high-dimensional data so that linear boundaries can separate the classes." (Chollet)

"Learning a representation means finding a transformation of the data such that the task at hand becomes easy to solve—often by making the relevant underlying manifolds linearly separable." (Bangio)

"The main goal of feature extraction and representation learning is to map complex, non-linearly separable input data into a feature space where simple, often linear, decision boundaries suffice." (Goodfellow)

"A good representation is one that makes the subsequent learning task easy. In classification, a good representation transforms a convoluted, non-linear decision surface into a simple hyperplane." (Bishop)

Course Overview

- We are not teaching common classifiers. They are taught in other courses (e.g., Machine Learning). We only use them as examples; you have to learn them yourself if you don't know them already.
- We are not teaching common feature extractors (such as for images).
- **We focus on understanding the feature space:**
  - How to construct it;
  - How to measure it;
  - How to manipulate it.
- This will become the general foundation for your understanding, regardless the kind of models or features in your tasks.
- As an overhauled curriculum, the weekly schedule is only approximate.

Course Layout (1)

- Classification overview: An quick overview of concepts and evaluation so that you can do the assignments.
- Linear spaces and metric engineering:
  - Inner products as the core.
  - Metric engineering, covariance geometry, Mohalanobis distance
  - Data whitening, PCA
  - Linear separability, LDA
  - Understanding curse of dimensionality
  - Linear classifiers and loss functions in feature space

Course Layout (2)

- Nonlinear spaces and manifolds:
  - Manifold assumption, intrinsic dimension
  - Manifold unrolling
  - Kernel framework
- Structural hypotheses in spaces
  - Clustering as test of structural hypotheses
- Representation learning and foundation spaces
  - Information theory basics
  - Self-supervised learning
  - Auditing foundation spaces

About This Course

- Instructor contact information:
  - Office: EC709
  - Office hour: By appointment
  - Phone: 56689
  - E-mail: wangts@cs.nycu.edu.tw
- Course web site: Use **New E3**
- Requirements
  - (12% x 3) Individual programming assignments + written report
  - (12% x 3) Quizzes (from lectures and assignments)
  - (28% x 1) Final exam

About This Course

- Policies:
  - Exam / quizzes are all closed book.
  - You're free to use toolboxes, open-source codes, or AI for the assignments. Just be responsible for what you submit.
- Grading:
  - Median semester grade will be around 84-86 out of 100 (can be higher if you all do very well).
  - For letter grades, generally I will allow up to 50~60% of the class in A+/A (can be more if you all do very well).
  - I will not accept requests for "doing extra work for better grade", including for school application or passing qualify exams.